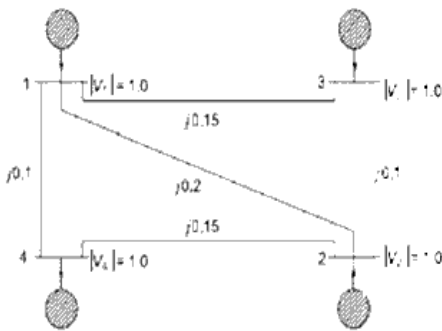
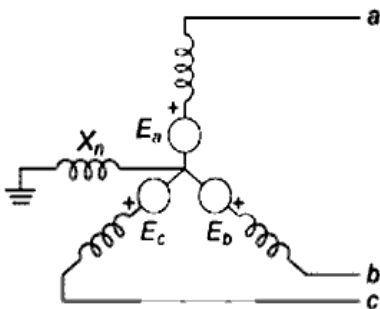


	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination Summer 2024 Course: B. Tech. Semester: V Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering. Subject Code & Name: BTEEC501 Power System Analysis Max Marks: 60 Date: 01/07/2024 Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. Use of non-programmable scientific calculators is allowed. 3. Assume suitable data wherever necessary and mention it clearly.	
		Marks
Q. 1	Attempt any two (2) of the following.	12
A)	Draw and explain impedance and reactance diagram.	6
B)	Define the following terms with suitable example: (i) Tree (ii) Branches (iii) Links (iv) Co-Tree (v) Basic loop	6
C)	Discuss per unit representation of single-phase transformer.	6
Q.2	Attempt any two (2) of the following.	12
A)	Explain positive sequence impedance and network of synchronous machine.	6
B)	Determine the symmetrical components of three currents $I_A = 10 \angle 30^\circ$, $I_B = 15 \angle -60^\circ$, $I_C = 18 \angle -154^\circ$	6
C)	Derive the bus admittance matrix by singular transformation	6
Q. 3	Attempt any two (2) of the following.	12
A)	Draw and derive an expression of short circuit transient current on transmission line.	6
B)	Derive an expression for fault current in double line to ground fault on unloaded generator. Draw sequence networks for fault current.	6
C)	Calculate the Y_{bus} matrix using Direct method. All line reactances are indicated in pu.	6

		
Q.4	Attempt any two (2) of the following.	12
A)	Discuss briefly the types of buses and explain the importance of load flow studies in power system.	6
B)	Explain Gauss-Seidel method of load flow solution.	6
C)	Draw and explain Automatic generation control system	6
Q. 5	Attempt any two (2) of the following.	12
A)	In Fig. a synchronous generator whose neutral is grounded through a reactance X_n. The generator has balanced emfs and sequence reactances X_1, X_2 and X_0 such that $X_1 = X_2 \gg X_0$. (a) Draw the sequence networks of the generator as seen from the terminals. (b) Derive expression for fault current for a solid line-to-ground fault on phase a. 	6
B)	Write a short note on Load Dispatch Centre (LDC)	6
C)	Discuss contingency analysis in power system.	6
	*** End ***	